CS 442
Introduction to Cryptography

Lecture 21: Public-Key Encryption

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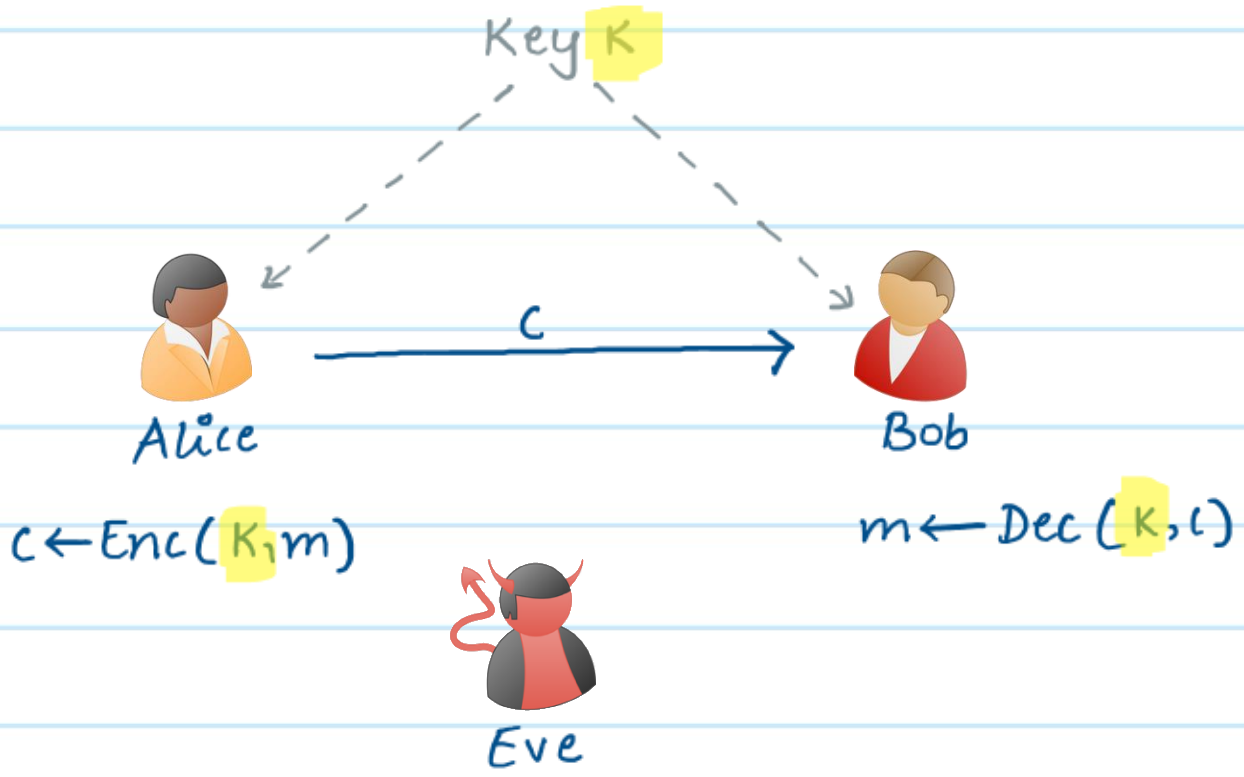
Agenda

* Public-Key Encryption (PKE)
* One-message secure PKE
* Applications of PKE
* ElGammal Encryption
* Multi-message secure PKE

* HW4 is due on April 19.

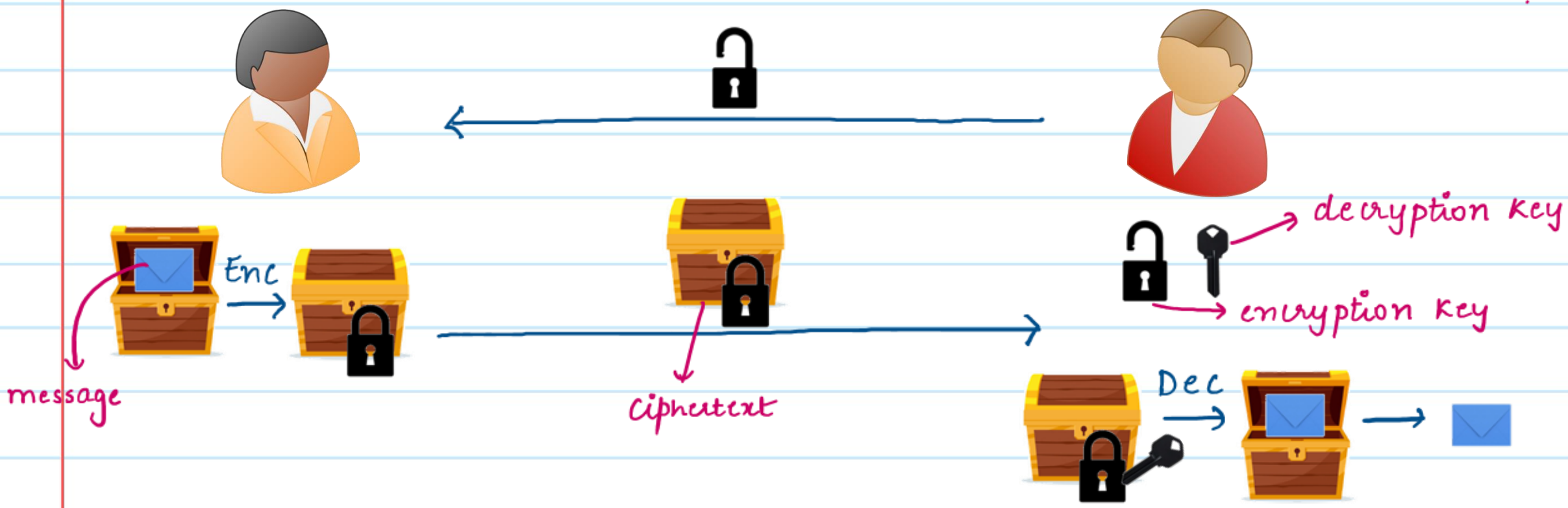
Secret-Key Encryption (SKE)

* So far, when discussing encryption schemes, we assumed that Alice and Bob use the same *secret-key* for both encryption and decryption.
* Such schemes are called secret-key encryption or symmetric-key encryption.



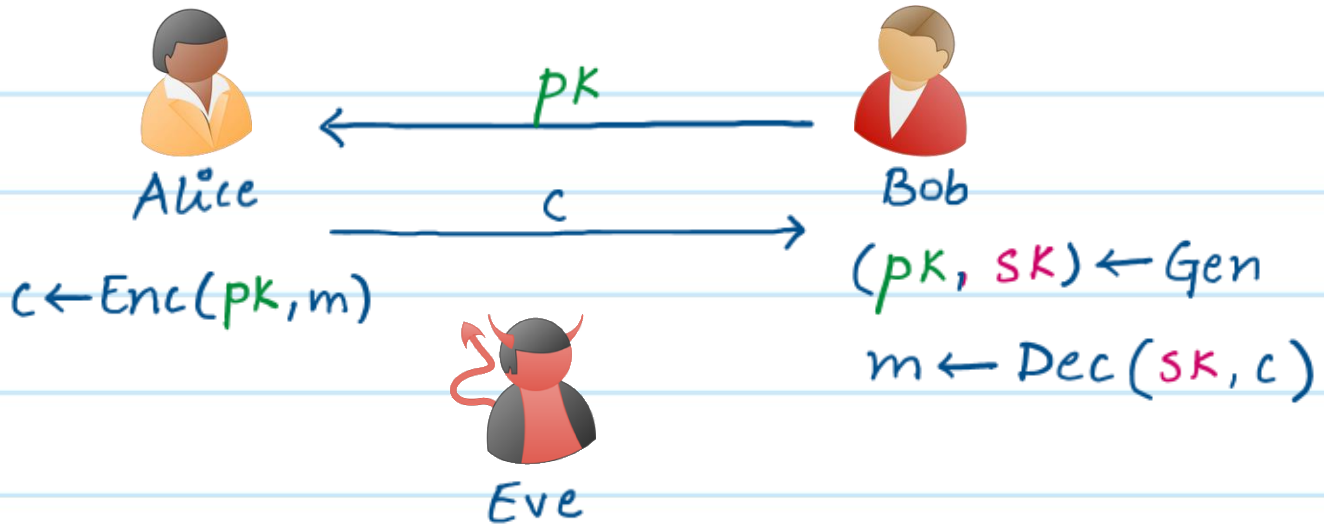
Public-Key Encryption (PKE)

* Public-Key encryption is a different type of encryption, where the key used for encryption is different from the key used for decryption.



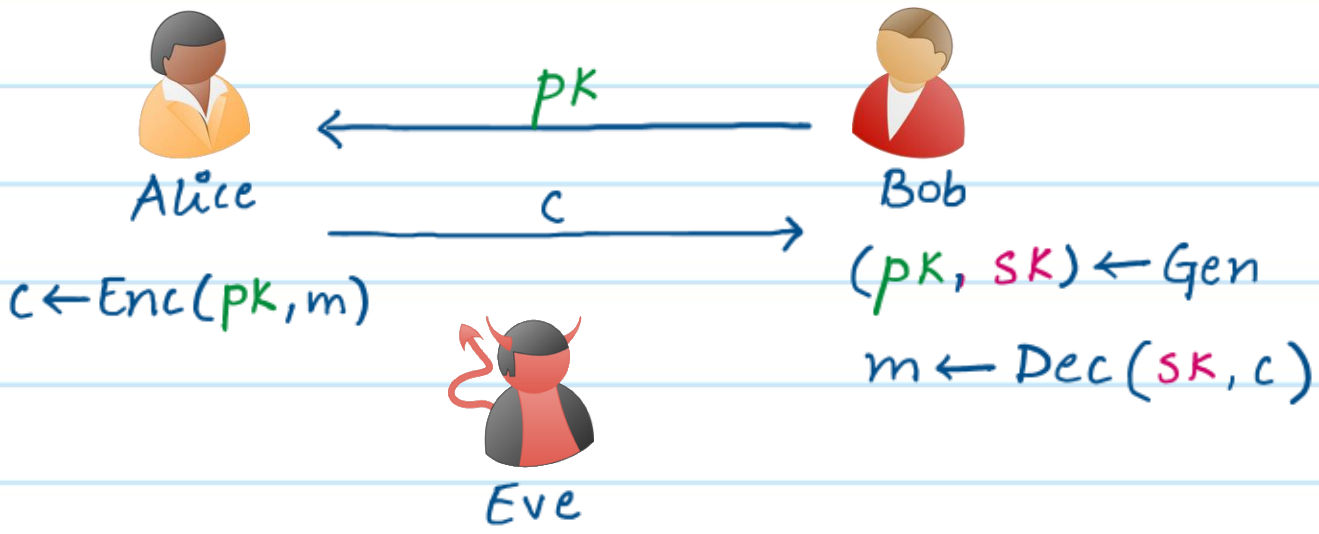
Public-Key Encryption

* Encryption Key = public Key (pk)
* Decryption Key = secret Key (sk)
* If Bob wants to receive a private message from Alice, he can generate a (pk, sk) pair and send pk to Alice.
* Alice uses pk to encrypt the message
* Bob uses sk to decrypt the message



Public-Key Encryption

* We don't need to hide the public key. It can be sent over a public communication channel. We only care about hiding the secret-key.
* Public-key does not reveal any information about the secret-key.
* No need for a key-exchange protocol.



Public-Key Encryption in the Real-World

* End-to-end encrypted messaging

* HTTPS (hypertext transfer protocol secure)

* VPN (Virtual Private Network)

One-Message Secure Public-Key Encryption

Definition: A one-message secure PKE scheme with message space $\mathcal{M}$, key space $\mathcal{K}$, ciphertext space $\mathcal{C}$, comprises of the following algorithms:

* $\text{Gen} \rightarrow pk, sk$: This algorithm samples a pair $(pk, sk) \in \mathcal{K}$.
* $\text{Enc}(pk, m) \rightarrow c$: On input a public key pk & a message $m \in \mathcal{M}$, it outputs ciphertext $c \in \mathcal{C}$
* $\text{Dec}(sk, c) \rightarrow m$: On input a secret key sk & a ciphertext $c \in \mathcal{C}$, it outputs message $m \in \mathcal{M}$.

These algorithms must satisfy the following:

→ **Correctness:** For $(pk, sk) \leftarrow \text{Gen}$, $\forall m \in \mathcal{M}$, the following holds:

$$\Pr[\text{Dec}(sk, \text{Enc}(pk, m)) = m] = 1$$

→ **Security:** For all PPT adversaries A, $\exists$ a negligible function $\nu(.)$, such that:

$$\Pr\left[\begin{array}{l} (pk, sk) \leftarrow \text{Gen}(1^n), \\ (m_0, m_1) \leftarrow A(pk), \\ b \xleftarrow{\$} \{0,1\} \end{array} \middle| \; A(pk, \text{Enc}(sk, m_b)) = b \right] \leq \frac{1}{2} + \nu(n).$$

Another Interpretation of One-message Security

Adversary

Challenger

$\xleftarrow{\quad pk \quad}$ $\quad (pk, sk) \leftarrow Gen(1^n)$

$\xrightarrow{\quad m_0, m_1 \quad}$ $\quad b \xleftarrow{\$} \{0,1\}$

$\xleftarrow{\quad c \quad}$ $\quad c = Enc(pk, m_b)$

$\xrightarrow{\quad b' \quad}$

$$\Pr[b = b'] \leq \frac{1}{2} + \nu(n).$$

OR equivalently: $\{pk, Enc(pk, m_0)\} \approx_c \{pk, Enc(pk, m_1)\}$

$\forall\, m_0, m_1 \in \mathcal{M}.$

Diffie - Hellman Assumptions

Let's sample a cyclic group $(\mathbb{G}, \cdot)$ of order q with generator g.

* Decisional Diffie-Hellman (DDH) Assumption: Sample $x, y, z \xleftarrow{\$} \mathbb{Z}_q$, compute $h_1 = g^x$, $h_2 = g^y$. Given $(\mathbb{G}, q, g, h_1, h_2)$, it is computationally hard to distinguish between g^{xy} and g^z.

In other words,

$$(\mathbb{G}, g, q, h_1, h_2, g^{xy}) \underset{\downarrow}{\approx_c} (\mathbb{G}, g, q, h_1, h_2, g^z)$$

Computationally indistinguishable.

ElGamal Encryption

Public-Key encryption for encrypting $m \in \mathbb{G}$.

* Gen: Sample a cyclic group $(\mathbb{G}, \cdot)$ of order q with generator g.
  Sample $x \xleftarrow{\$} \mathbb{Z}_q$. $pk = (\mathbb{G}, g, q, X = g^x)$, $sk = (\mathbb{G}, g, q, x)$

* Enc(pk, m): Sample $y \xleftarrow{\$} \mathbb{Z}_q$. $c = (Y = g^y, D = m \cdot X^y)$

* Dec(sk, c): Parse $c = (Y, D)$. Compute $m = D \cdot Y^{-x}$.

* **Correctness:** We need to show $\Pr[\mathrm{Dec}(sk, \mathrm{Enc}(pk, m)) = m] = 1$.

$\mathrm{Enc}(pk, m) = (Y = g^y,\ D = m.X^y)$

$\mathrm{Dec}(sk, c)$: $\quad D \cdot Y^{-x} = m.X^y.Y^{-x} = m.g^{xy}.g^{-xy} = m$

* **One-message Security:** We need to show:

$$\forall\, m_0, m_1 \in \mathcal{M} : \{pk, \mathrm{Enc}(pk, m_0)\} \approx_c \{pk, \mathrm{Enc}(pk, m_1)\}$$

$$\{\mathbb{G}, g, q, g^x, g^y, m_0 \cdot g^{xy}\} \qquad \{\mathbb{G}, g, q, g^x, g^y, m_1 \cdot g^{xy}\}$$

Lets use hybrid arguments.

H_0: $\{\mathbb{G}, g, q, g^x, g^y, m_0 \cdot g^{xy}\}$

H_1: $\{\mathbb{G}, g, q, g^x, g^y, m_0 \cdot g^z\}$ random $z \xleftarrow{\$} \mathbb{Z}_q$

H_2: $\{\mathbb{G}, g, q, g^x, g^y, m_1 \cdot g^z\}$

H_3: $\{\mathbb{G}, g, q, g^x, g^y, m_1 \cdot g^{xy}\}$

* $H_0 \approx_c H_1$: follows from the DDH assumption
* $H_1 \equiv H_2$: perfectly indistinguishable (like one-time pad)
* $H_2 \approx_c H_3$: follows from the DDH assumption

From hybrid lemma, it follows that $H_0 \approx_c H_3$.

PKE Cannot be Deterministic

* The encryption algorithm in a PKE scheme must be a randomized algorithm, in other words, it must take an additional random value as input.

WHY?

* If the encryption algorithm is deterministic, then given a pk, there will be a unique ciphertext corresponding to each message.
* A PPT adversary in this case has pk and can encrypt both m_0 and m_1 on his own, compare these to the ciphertext sent by the challenger to deduce whether its an encryption of m_0 or m_1 in the one-message security game.

Perfectly Secret PKE is Impossible.

* No PKE scheme can be secure against a computationally unbounded adversary.

WHY?

* We have already established that the encryption algorithm in a PKE scheme must be randomized.
* An unbounded adversary can easily win the one-message security game by computing all possible encryptions of m_0 and m_1 using all possible random values. The adversary can then check whether the ciphertext received from the challenger corresponds to one of the encryptions of m_0 or m_1.

Multi-Message Security

Adversary

Challenger

$\xleftarrow{\quad pk \quad}$	$(pk, sk) \leftarrow Gen(1^n)$
$\xrightarrow{\quad m_0^1,\ m_1^1 \quad}$	$b \xleftarrow{\$} \{0,1\}$
$\xleftarrow{\quad c^1 \quad}$	$c^1 = Enc(pk, m_b^1)$
$\xrightarrow{\quad m_0^2,\ m_1^2 \quad}$	$c^2 = Enc(pk, m_b^2)$
$\xleftarrow{\quad c^2 \quad}$	$\vdots$
$\vdots$	
$\xrightarrow{\quad m_0^k,\ m_1^k \quad}$	$c^k = Enc(pk, m_b^k)$
$\xleftarrow{\quad c^k \quad}$	
$\xrightarrow{\quad b' \quad}$	

$Pr[b = b'] \leq \nu(n)$

Multi-Message Security

* Every one-message secure PKE is also multi-message secure.

WHY?